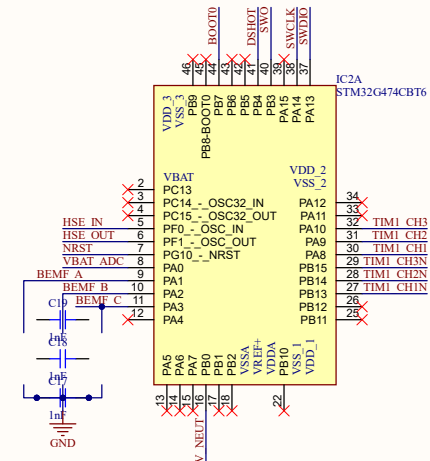


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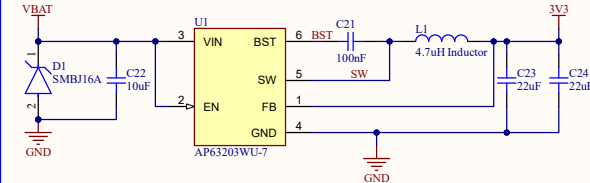
Wiz ESC

-Custom sensorless electronic speed controller
-Nominally takes a 3S battery but can take up to a 4S

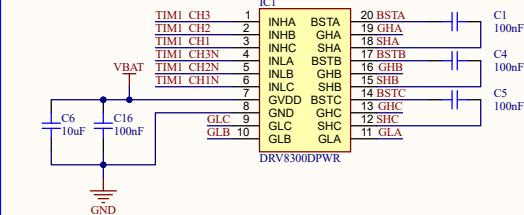
STM32G4 MCU



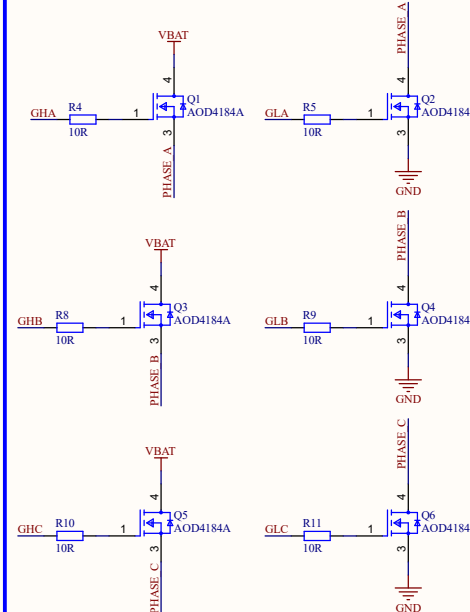
VBAT to 3V3 and protection



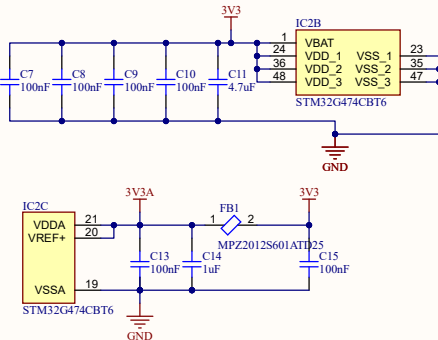
Gate Driver and circuitry



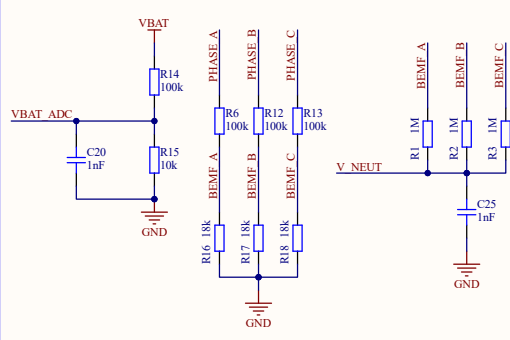
Mosfets and phase outputs



STM decoupling and analog



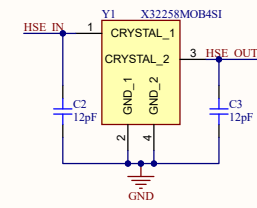
Voltage sensing for VBAT, back emf and neutral point



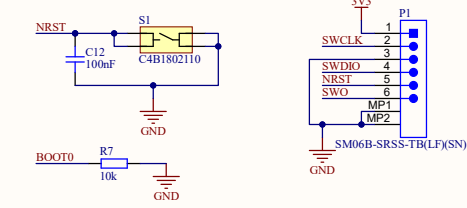
Notes:

- Values for resistors and capacitors have largely been chose based of datasheet recommendations.
- Except for C_GVDD, C_BSTx and R_GHx/GLx on the gate driver/mosfet gates.
- Voltage divider resistances have been decided on to ensure no damage is done to the MCU ADC pins and also allow for relatively accurate readings.
- Other valeus not mentioned in any datasheets, e.g. 1nF decoupling on the ADC lines has been chosen out of respect for the nature of their purpose.
- Generally parts chosen for simplicity, good documentation and overall fitting the needs.

8MHz Crystal for STM



Boot, Reset and SWD for STM



APPROVALS	DATE	PROJECT
ENG: Barney		
DSN: Barney		
CHK: Barney		
REFERENCE DOCUMENTS	FILE NAME	DATE
ASSY DWG:	ESC_Main.SchDoc	2024-05-28
FAB DWG:		
PCB DWG:		
SIZE B	CAGE CODE	DWG NO.
SCALE:	FILE NAME	SHEET 1 OF 1

Altium

PROJECT REVISION: 101e704106762d016ccda38ed3a2c9a3b2b0d24d528a369

DOCUMENT REVISION: 101e704106762d016ccda38ed3a2c9a3b2b0d24d528a369

DESIGN ITEM: 101e704106762d016ccda38ed3a2c9a3b2b0d24d528a369

TITLE: Wiz ESC

REV

REV

REV